



MILITARY INSTITUTE OF SCIENCE AND TECHNOLOGY

Project Proposal

PitchSync: A Centralized Cricket Board Management and Integrity Monitoring System

CSE-302: Database Management System Sessional

Group – A9

Section-A

Department of Computer Science and Engineering

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1. Introduction

PitchSync is a centralized cricket board management and integrity-monitoring system primarily designed from the operational perspective of the Bangladesh Cricket Board. Although the initial design context is Bangladesh, the database follows a generalized cricket administration model and can be adapted by other national or regional cricket boards. The project focuses on the structured flow, storage, retrieval, and management of cricket-related data within an internal board environment.

The system integrates player information, team records, tournament and match administration, match-summary performance statistics, career records, player fitness and availability, complaints, persons of interest, rulebook enforcement, and case investigation in a unified database. It supports both men's and women's cricket and organizes records across international, domestic, franchise or league, and youth or academy competition tiers. Supported match formats include Test, One Day/ODI, T20, First-Class, List A, and Domestic T20.

PitchSync records match-level batting, bowling, and fielding summaries and accumulates them into career records according to competition tier and match format. The platform also enables authorized officials to record complaints received through different channels, identify relevant persons of interest, register disciplinary or legal cases, connect cases with applicable rulebook clauses, and track whether a matter remains internal or is referred to external law-enforcement authorities.

The platform is intended strictly for internal cricket board operations. It does not provide public or guest access, automated legal judgments, full clinical medical-record management, or ball-by-ball scoring. Player injury status, recovery progress, fitness clearance, and match or training availability are maintained only for operational team-management and selection purposes.

2. Project Objective

- To develop a centralized Oracle-based database for storing and managing cricket board information in a structured and consistent manner.
- To maintain complete player profiles, team histories, competition records, match-summary performances, and career statistics for both men's and women's cricket.
- To assist team officials in monitoring player injury status, recovery progress, fitness clearance, and availability for training or matches.
- To support tournament, match, team, and player-performance administration across multiple competition tiers and match formats.
- To provide role-based access so that each internal user can view or modify only the information relevant to assigned responsibilities.
- To record complaints and suspicious activities and support the identification and monitoring of persons of interest.
- To register, update, and trace internal disciplinary cases and cases referred to external law-enforcement authorities.
- To connect cases with complaints and applicable rulebook clauses, thereby improving transparency, accountability, and evidence-based decision-making.

3. Users

Super Administrator: Controls the complete system, manages user accounts, assigns roles and permissions, and maintains system-level access and security settings.

Cricket Board Administrator: Maintains player, team, tournament, match, and general administrative records and monitors overall cricket board operations.

Team Performance Manager: Combines team-management and coaching responsibilities by reviewing player profiles, fitness and availability, career records, and match performances to support selection and team planning.

Match Official or Scorer: Records match details, results, and final batting, bowling, and fielding summaries and verifies match-level performance data.

Integrity and Compliance Officer: Records complaints, identifies persons of interest, opens and investigates cases, links cases to rulebook clauses, tracks referrals, and updates disciplinary or legal case status.

Player: Has limited access to view personal profile, fitness and availability information, match performances, and career records without modifying official administrative, performance, or case data.

4. Front End and Back End Tools

Category	Technology	Purpose
Database	Oracle Database 12c (Fully backward compatible with Oracle 11g)	Stores relational data, constraints, relationships, queries, and transaction records.
Application and Backend	Next.js (App Router)	Provides application routing, server-side functionality, and backend request handling.
Application and Backend	Node.js	Supports server-side execution and communication between the application and database.
Frontend / UI	React	Builds reusable and interactive user-interface components.
Frontend / UI	Tailwind CSS	Provides responsive and consistent interface styling.
Frontend / UI	Framer Motion	Adds limited, smooth interface animations and transitions.

5. Features of Project

1. User and Role Management: Creates and manages internal accounts for the six approved user roles and applies role-based access to sensitive sporting, administrative, fitness, and disciplinary information.

2. Player Profile, Fitness and Availability Management: Stores player identity, playing role, address, educational and family background, gender category, injury status, recovery progress, expected return date, fitness clearance, and training or match availability. It does not store full clinical records or provide medical diagnosis.

- 3. Team Management:** Maintains team names, categories, franchise ownership where applicable, squad information, and the history of teams represented by each player.
- 4. Tournament and Match Management:** Manages tournaments, participating teams, matches, dates, venues, competition tiers, match formats, results, and match status for men's and women's cricket.
- 5. Player Performance Management:** Stores final match-summary batting, bowling, and fielding records. Batting data includes runs, balls faced, dismissal type, and strike rate; bowling data includes wickets, overs, runs conceded, and economy rate; fielding data includes catches, stumpings, direct run-outs, and byes conceded. Ball-by-ball records are outside the project scope.
- 6. Career Record Management:** Accumulates match-level performance into batting, bowling, and fielding career summaries according to competition tier, match format, team, and number of matches played.
- 7. Complaint and Integrity Monitoring:** Allows complaints from different sources to be recorded only by an authorized Integrity and Compliance Officer, including description, submission date, source, and suspected misconduct. Relevant players, officials, team personnel, franchise personnel, support staff, or other cricket-related individuals may be designated as persons of interest.
- 8. Case and Rulebook Management:** Registers one or more cases from a complaint or suspicious activity, records opening date, status, involvement type, and referral status, and connects each case with relevant persons, complaints, and allegedly violated rulebook clauses. The system supports both internal disciplinary cases and matters referred to external law-enforcement authorities.
- 9. Reporting and Monitoring:** Provides authorized users with searchable summaries and consolidated views of player careers, team and match records, fitness and availability, complaints, persons of interest, rule violations, and case progress for operational monitoring and decision-making.